

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Chemistry

2. Subject Name: Molecular Thermodynamics and Molecular Reaction Dynamics
L-T-P: 3-1-0 Credits:4

3. Pre-requisites: Introduction to Quantum Chemistry & Spectroscopy

4. Syllabus and reference books:

Syllabus:

Module 1: Molecular Thermodynamics

1. Review of probability; distribution functions
2. Introduction to statistical thermodynamics; thermodynamic states of macroscopic systems
3. Microstates of simple systems – definition and example in exactly solvable systems including two-state systems, simple harmonic oscillators, particle in a box and rigid rotors; concept of ensembles
4. Microcanonical ensemble - Boltzmann hypothesis and hypothesis of equal a priori probability, entropy in exactly solvable systems
5. Canonical ensemble – partition function; thermodynamic quantities from partition functions in exactly solvable systems; fluctuations of energy
6. Quantum statistics of identical particles– Fermi-Dirac and Bose-Einstein statistics; relation to Maxwell-Boltzmann distribution
7. Applications to ideal gases – effect of atomic structure on the thermodynamics of monatomic ideal gases; effect of structure on specific heat of heteronuclear diatomic ideal gas
8. Chemical equilibrium in the gas phase – effect of molecular structure on equilibrium constant; molecular basis of the le Chatelier's principle
9. Normal modes of vibration and vibrational contribution to specific heat of solids – Einstein model
10. Introduction to classical statistical mechanics
11. Inter-molecular interactions and application to equation of state of real gases
12. Radial distribution function: structure and thermodynamics of atomic liquids
13. Polymers and biomolecules – application of the concept of two state molecular thermodynamics to topics like conformations, ligand binding and folding

Module 2: Molecular Kinetics and Reaction Dynamics

1. Rates of Chemical reactions: Enzyme Catalysis, Autocatalysis, Exact Analytic solution for Complex Reactions
2. Kinetic Measurements: Techniques for Kinetic Measurements (Flow Technique, Relaxation Method, Flash Photolysis and laser flash photolysis, Ultrafast Laser techniques)
3. Reaction in Solutions: General properties of reactions in Solution, Phenomenological Theory of Reaction Rates, Diffusion-Limited Rate Constant, slow reactions, Effect of ionic strength on Reaction between ions, Linear Free-Energy Relationship, Electron

Transfer Reactions: Marcus theory	
4.	Theories of Chemical Reactions: Potential energy Surfaces of Chemical Reactions, Collision Theory, Transition State / Activated Complex Theory: Basic Postulates and Derivation of the Transitions State Theory (TST), Quantum Mechanical Effects in TST, Thermodynamic Formulation of TST, Applications of TST, Symmetry and Statistical factor
5.	Transport Properties: Functional form of the Transport equations, Time dependent; transport; Langevin equation and fluctuation-dissipation theorem
Reference Books:	
1)	Physical Chemistry Statistical Mechanics, H. Metiu, Taylor & Francis, 2018.
2)	Molecular Thermodynamics, D. A. McQuarrie and J. D. Simon, Viva Books, 2016.
3)	Chemical Kinetics and Dynamics, J. I. Steinfeld, J. S. Francisco and W. L. Hase, Prentice Hall International, 1999.
4)	Chemical Kinetics and Reaction Dynamics, P. L. Houston, Dover Publications, 2001.
5)	Chemical Kinetics, K. J. Laidler, Pearson Education (Indian reprint) 2004.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Module 1: Molecular Thermodynamics	18
2.	Module 2: Molecular Kinetics and Reaction Dynamics	18
3.	Tutorial	12
Total number of hours		48